Preparing Your Data: Scaling & Splitting

GeoPython — Quick Reference Handout 03

Linah Ababneh, Ph.D.
LA Research & Consulting

1. Why Scale?

Machine learning algorithms that rely on distances — K-Nearest Neighbors (KNN), Support Vector Machines (SVM), gradient descent-based models — treat all numbers as literal magnitudes. A variable measured in millions will dominate a variable measured in single digits, distorting the model's attention.

Variable	Raw Range	After Scaling
Sea Ice Extent	0 – 14,000,000 km²	0.0 – 1.0
Temperature Anomaly	-3.5 – +4.2 °C	0.0 – 1.0
Solar Zenith Angle	0° – 90°	0.0 – 1.0

! *Scaling does not change your data — it changes the units of comparison. The relationships between values are preserved; only the scale changes.*

2 . Common Scaling Methods

Method	Formula (concept)	When to Use
Min-Max (Normalisation)	(x − min) / (max − min)	When you need values in [0, 1]; sensitive to outliers
StandardScaler (Z-score)	(x − mean) / std	When distribution is roughly normal; default choice
RobustScaler	(x − median) / IQR	When outliers are common (e.g., storm events)

std = standard deviation (how spread out values are around the mean). IQR = Interquartile Range (distance between the 25th and 75th percentile; resistant to extreme values).

3 . The Seasonal Train/Test Split

A random 80/20 split can be dangerous in environmental science. If your data is seasonal — e.g., the ice sheet dataset explored in the companion notebook — a random split mixes winter and summer records, and the model may then predict summer conditions using winter patterns, inflating apparent accuracy.

Split Strategy	Risk Level	When Appropriate
Random 80/20	⚠ High (seasonal data)	Only if data has no temporal structure
Chronological split	✓ Lower	Time-ordered data; test on future dates
Season-filtered split	✓ Best for seasonal/interval data	Any data with a cyclical or interval structure — climate & weather, streamflow & water runoff, annual growth rates, phenological cycles

👁 *Filter for your target season BEFORE splitting. This ensures training and test sets contain only the patterns the model is supposed to learn — no cross-season contamination.*